

QBUS2820 Predictive Analytics, S2 2026

Week 1: Introduction to Predictive Modelling

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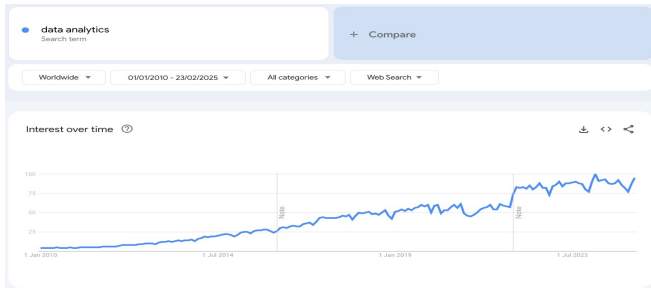
Week 1: Introduction to Predictive Modelling

1. Introduction
2. What is QBUS2820 about?
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Introduction

A big picture of business analytics process

- "The world's most valuable resource is no longer oil, but data" - *The Economist*
- It's perhaps not too exaggerated to say that if a business has access to the necessary data and is able to convert the data into business insights, it can conquer the business world.
- Below is Google search trend of the term "Data analytics".
What does this tell you?



A big picture of business analytics process

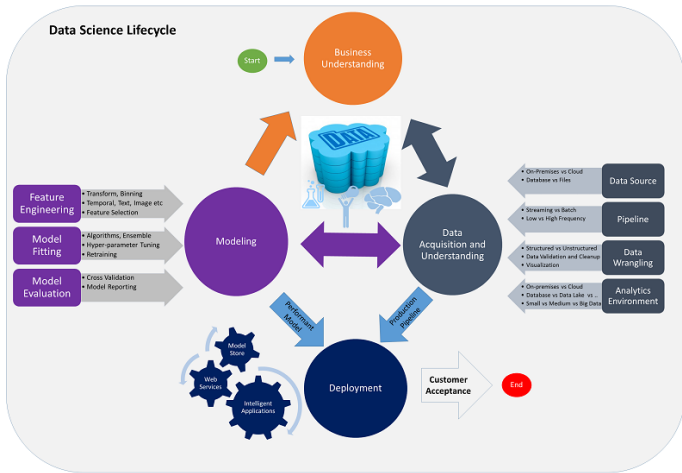
Start with a business problem thought to be solvable using data analytics, e.g.,

- Why have the sales decreased in the last two quarters?
- A number of the bank's ATMs were cash-overloaded, can we predict the ATM cash-demand better?
- Can we detect potentially fraudulent bank transactions?

How and where should you start?

Note that not all business problems can be solved using data analytics, but some can.

A big picture of business analytics process



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¹<https://docs.microsoft.com/en-us/azure/machine-learning/data-science-process-overview>

Business Analytics process

- **Business understanding:** Know the business context and what you're dealing with. Raise the right business question. Turn the business problem into a data analysis problem.
- **Data acquisition and understanding:** Understand the appropriate data needed for the given business objectives, how/where to get the proper data.
- **Modelling:** Develop a mathematically sound framework to achieve the business objectives outlined during the business understanding phase. This step is crucial and involves significant mathematical challenges.
- **Deployment:** Implement the data analytic findings. The insights and models developed are handed over to software engineers and relevant business teams for integration into the business processes and systems.

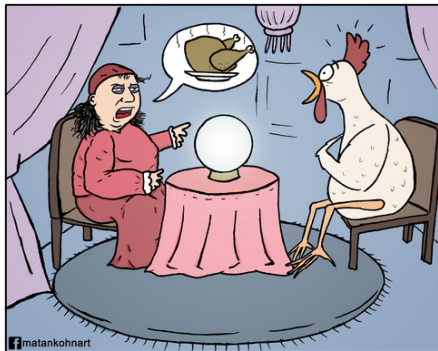
What are covered in this unit?

- This unit focuses mainly on the **Modelling** step, with some demonstration of the entire BA process as much as possible
- HUGE number of data analysis techniques can be useful in Modelling: inference, prediction, regression, classification, deep learning, Bayesian computation, etc. This unit only covers a tiny part of this. *Yes, knowledge is infinite!*
- More precisely, this unit focuses on *Predictive modelling*

Predictive modelling is a set of methods for detecting patterns in data and using these patterns for predicting future data and informing decision making. It draws on methods from the fields of statistics, econometrics, and machine learning.

What are covered in this unit?

You might become a *data-based* fortune teller after studying this course²



²Picture credit:

https://www.toonpool.com/cartoons/Vegetarianism_283755

What is QBUS2820 about?

QBUS2820 content structure

1. Statistical Machine Learning foundations and applications: key concepts in predictive modelling, statistical thinking, model evaluation, model selection, and model inference, etc.
2. Regression: LR, KNN, ridge regression, LASSO, etc.
3. Classification: KNN, logistic regression, regularized logistic regression, etc.
4. Forecasting: key concepts, time series, exponential smoothing and ARIMA models.

Learning outcomes

By successfully completing this unit, you are expected to:

1. Understand the conceptual and **theoretical** foundations of predictive modelling.
2. Develop an in-depth knowledge of basic **methods for regression, classification, and forecasting** methods for business applications.
3. Be able to conduct **a complete data analysis project** based on these foundations and methods.
4. Know how to use **Python** for your practical workflow under realistic data complexity (including tasks such as data manipulation and visualization).
5. Effectively communicate your results to guide decision-making.

Comments

- This unit is designed as training for real-world predictive analytics, which requires a range of skills.
- Practical work in this area involves more than knowing the methods in the lectures: professionals typically spend a substantial amount of time on tasks such as data management, exploratory data analysis, feature engineering, and implementing methods.
- All of this is generally done through coding. Therefore, Python is your bridge between knowledge and practice.
- For these reasons, please note that this unit requires independent work and higher-than-average workload (within the university guidelines).

Admin stuff

Teaching Material

- Comprehensive lecture slides will be given weekly
- Information on the Canvas site
<https://canvas.sydney.edu.au>.
- Tutorial material
- Some materials I will give during the course which may be of interest to you

Check the Canvas site at least twice a week! Lectures, tutorials, related info are constantly posted there.

Suggested textbooks

ISL *An introduction to statistical learning: With applications in R.* by G James et al. Springer-Verlag: easy to read, sloppy sometimes, come with R/Python code for practice.

ESL *The Elements of Statistical Learning* by Hastie et al.
<https://web.stanford.edu/~hastie/ElemStatLearn/>:
theory-oriented version of ISL, well-written, deep in theory, suitable for students with a sound maths background.

FPP *Forecasting: principles and practice.* Rob J Hyndman and George Athanasopoulos. Open access book available at www.otexts.org/fpp2/

The Software for the Unit

- Python
 - Free and Works on PCs, Mac, Unix/Linux
 - Does statistical modelling, visualisation and programming environment
 - Can be used for almost all models to be discussed in this class
- Matlab and R are also very good for data analysis, but not used in this unit.
- Excel/SPSS are not enough to complete most data analysis tasks in this course

The teaching team

- Coordinator: Qin Fang
Webpage: <https://qinnnnnf.github.io/>
Email: qin.fang@sydney.edu.au.
- Tutors:
 - Emma Nguyen, Viet Hung Dao and Kuan (Michael) Yan
- See [Canvas](#) for the tutors' details, tutorial timetable, consultation times, etc

Assessments

	Release	Due
Assignment 0 – EFT (10%)	Week 1	17/08/2025 (Week 3)
Assignment 1 – Regression (25%)	Week 4	25/09/2025 (Week 8)
Assignment 2 – Time series (25%)	Week 10	06/11/2025 (Week 13)
Final exam (40%)		

- Assignment 0 (EFT) is due **23:59, Monday, 17th August**. This deadline is final. Late submissions will not be accepted, and any outstanding assignments will receive a mark of zero.
- The marks for Assignment 2 will NOT be released before the final exam.

Communication

- Talk to the teaching team [during or after classes](#), or at [consultation times](#)
- Post questions on Ed Forum
 - Before you ask PYTHON questions here, please try asking the [virtual 2820 python helper](#) first.
 - Anyone can see and contribute to the discussion. There's also an option for private matters.
 - Average response time: 3 working days.
 - No monitoring on weekends.
 - For heavy-technical questions, best to ask at consultation times
- Using personal emails is the least preferred option.

Some Tips

- Learning by doing;
- Ask questions;
- We focus on technical materials, but remember the profile of a good business analyst: common sense, creation, communication and data visualization skills are very important.

Let's start our predictive analytics
journey...

Some key concepts and notations

Types of prediction

Different types of data lead to different types of prediction problems:

- In **cross-sectional prediction**, we work with data all collected at a particular time point. E.g., for house value evaluation, we observe several variables such as selling price, location, number of rooms, size, etc. for many sold houses across Sydney, at a particular time point (such as 2026). Our task is to predict the value of a house to be put on the market.
- In **time series forecasting**, we want to predict the value of a variable at a specific time point in the future, based on past and current information. Here, time at which we observed the past data, and at which we want to forecast, is important.

Introduction to statistical learning

- Q1: Supervised learning vs Unsupervised learning.
- Q2: Regression vs Classification.

Supervised learning

In the context of statistical learning, **supervised learning** is the task of learning a function to predict an output variable Y_0 based on observed input variables $\mathbf{x}_0 = (x_{01}, \dots, x_{0p})^T$. We develop methods that learn this function based on labelled data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, which we call the **training data**.

Supervised learning: Regression or Classification

In supervised learning, the output or **response variable** can be of any type. We will study methods that address two main classes of supervised learning problems:

- In **regression**, the response is a quantitative scalar (such as the income of a worker).
- In **classification**, the response is **nominal** or **categorical** variable $Y \in \{1, \dots, C\}$, where C is the number of classes. When $C = 2$, this is called binary classification; if $C > 2$, this is called multiclass classification. The response can be **ordinal** - but not covered in this unit.

Example: handwritten digit recognition



A view of the MNIST dataset.

Unsupervised learning

Unsupervised Learning: There are no input or output variables. The data are *unlabelled*. Learning tasks are to uncover hidden patterns, clusters, relationships between the variables or distribution of the data. Examples of unsupervised learning techniques:

- **Clustering** (also known as segmentation): cluster data into groups with similar characteristics.
- **Principle Component Analysis:** find a few key factors that determine the data patterns.

Notation

- We use upper case letters such as Y to denote random variables, regardless of dimension.
- Lower case letters denote observed values. For example, y denotes the realised value of the random variable Y .
- We use i to index the observations, j to index the inputs. For example, y_i is the observed response for sample i , while x_{ij} is the value of predictor j for observation i .
- We use the hat notation (e.g. $\hat{\beta}$) for estimators and estimates. The notation may not distinguish between the two (refer to context).
- Vectors are in lower case bold letters. Matrices are in upper case bold letters.

Vector and matrix notation

Response vector:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Please review “Maths for business” available on Canvas

Vector and matrix notation

Vector of predictor (features, attributes, covariates, regressors, independent variables) values for observation i :

$$\mathbf{x}_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}$$

Vector of observed values for predictor j :

$$\mathbf{x}^j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix}$$

Vector and matrix notation

Design matrix:

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}$$

From Data to Prediction: What Are We Really Learning?

Hot Chocolate Sales: A Winter Prediction Problem

Let's say you run a hot chocolate stall at the Winter Wonderland Festival, held every June–August.

Each day:

- X = daily temperature;
- Y = number of hot chocolates sold.

You've recorded this data from previous winters: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$.

Goal: Use the past data to help predict how many hot chocolates to prepare *tomorrow*, given the forecast temperature.

Where does the data come from?

- Each pair (x_i, y_i) is a data point — but where does it come from?
 - X (temperature) is a **random variable**: it changes from day to day, drawn from a winter temperature distribution.
 - Y (sales) is also **random**: even on the same X , people may buy more or fewer hot chocolates.
 - So the dataset is a **sample from an underlying joint distribution**.
- Your data is a sample — a snapshot of randomness.
- The distribution is the engine that generates it.

Our Goal: Learning the Data-Generating Process

- Sample vs. Distribution:
 - **Sample** (Dataset): A collection of observed values — for example, 20 days of (x_i, y_i) pairs such as temperature and hot chocolate sales. This is the data **we actually observe**.
 - **Population** (Distribution): The full (possibly unknown) process that generates the data. It captures the “secret engine” behind the data — and is **what we hope to understand or model**.

Takeaway: Our goal is to uncover the *data-generating process*.

However, this full distribution is often extremely hard or impossible to recover or estimate.

Formal Definition of Prediction

We define prediction as follow:

1. Use the training data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ to learn a predictive function $\hat{f}(\mathbf{x})$.
2. Given a new input $\mathbf{x}_0$, make the prediction $\hat{f}(\mathbf{x}_0)$.

Key questions:

- How should we approach the prediction task?
- How do we define an **objective** that is both meaningful and achievable?
- How can we measure **success** in reaching that objective?

To provide a framework (**intuition**) for developing statistical learning models, we now turn to **decision theory**. We first focus on regression problems for simplicity.

Objective: Loss function

A **loss function** or **cost function** $L(y, \hat{y})$ measures the cost of prediction $\hat{y}$ when the truth is y . The most common loss function for regression is the **squared loss**:

$$L(y, \hat{y}) = (y - \hat{y})^2$$

For binary classification, a typical loss function is the **0-1 loss**:

$$L(y, \hat{y}) = \begin{cases} 1 & \text{if } y \neq \hat{y} \\ 0 & \text{if } y = \hat{y}. \end{cases}$$

Expected (squared) prediction error

We now define the **expected loss** or **expected prediction error/risk**:

$$\text{EPE}(f) = E \left[\{Y - f(X)\}^2 \right],$$

where the expectation is over the joint probability distribution $P(X, Y)$. This quantity measures the average squared error of a given prediction function $f(\cdot)$.

Our goal is to find the prediction function $\delta(\cdot)$ that minimizes EPE. This is equivalent to minimising EPE at every input point $\mathbf{x}$:

$$\delta(\mathbf{x}) = \underset{f(\cdot)}{\operatorname{argmin}} E(\{Y - f(X)\}^2 | X = \mathbf{x})$$

Regression function

The solution is the conditional expectation:

$$\delta(\boldsymbol{x}) = E(Y|X = \boldsymbol{x})$$

Takeaway:

- The best prediction of Y at any point $X = \boldsymbol{x}$ is the conditional mean $E(Y|X = \boldsymbol{x})$, when the best is measured by the average squared error.
- Prediction isn't about memorizing data. It's about understanding how Y depends on X .

Statistical modelling

- Our regression problem reduces to the estimation of the regression function $E(Y|X = \boldsymbol{x})$ (often unknown).
- Estimating an unknown regression function requires assumptions, which lead to statistical models.
- These assumptions guide how you approach the problem and define the methods used to build and validate your model.
- For example, the linear regression model assumes that $E(Y|X = \boldsymbol{x})$ is linear in $\boldsymbol{x}$:

$$E(Y|X = \boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{\beta}$$

Additive error model

The **additive error model** is our basic general model for regression. It assumes that the relationship between Y and X is described as

$$Y = f(X) + \varepsilon,$$

where $f(\cdot)$ is an unknown **regression function**, and ε is a random error term, which is independent of X and has mean zero, i.e.

$$E(\varepsilon) = 0.$$

Under this model,

$$E(Y|X = \mathbf{x}) = E(f(\mathbf{x}) + \varepsilon) = f(\mathbf{x}),$$

since $E(\varepsilon) = 0$.

Example: linear regression

In the special case of the linear regression model, we assume that

$$f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p,$$

leading to the model

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon,$$

and predictions

$$\hat{f}(\mathbf{x}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p,$$

where $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ is the vector of least squares estimates of the model parameters.

Why estimate f : Prediction

- Focus: make **accurate and reliable forecasts** for unseen or future observations.
- Evaluation metrics: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), etc.
- Example: In predicting tomorrow's weather, the focus is on providing accurate forecasts of temperature, humidity, precipitation, etc., based on historical weather data.

Why estimate f : Inference

- Focus: understand and draw conclusions about the **underlying relationships or patterns** in the data.
 - *Which predictors are associated with the response?*
 - *What is the relationship between the response and each predictor?*
- Evaluation methods: hypothesis testing, confidence intervals, and assessing the significance of relationships.
- Example: In inferring the effect of a new drug on patients' health outcomes, the focus is on understanding whether the drug has a statistically significant impact and estimating the magnitude of this effect.

Summarize: Types of learning

- **Supervised learning:**
 - Based on the data type of Y :
 - **Regression**: Y is continuous/numerical.
 - **Classification**: Y is categorical.
 - Two reasons for estimating f :
 - **Prediction** (predictive accuracy).
 - **Inference** (association between Y and X).
- **Unsupervised learning.**

Statistical decision theory

Our discussion of statistical decision theory lays the foundation for the model training and model selection discussed later in the course.

- **Objective function** represents the metric to be optimized during the learning process. More loss functions will be introduced in later chapters.
- **Model specification** is the process of defining the structure and assumptions of a statistical model.

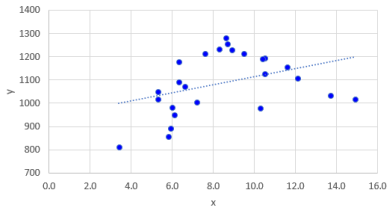
Together, they are essential for building effective and interpretable statistical models.

Other key concepts and themes

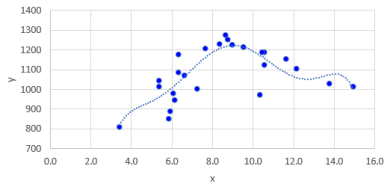
Other key concepts and themes

- Underfitting and Overfitting.
- Parametric vs non-parametric models.
- No-free lunch theorem.
- Accuracy vs interpretability.

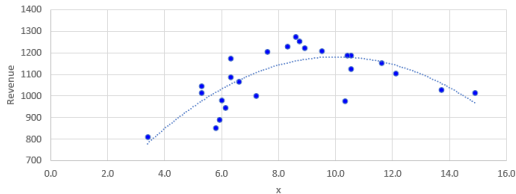
Underfitting and Overfitting



Underfitting: High bias, low variance



Overfitting: High variance, low bias



Best model

Overfitting

- We say that there is **overfitting** when an estimated model is excessively flexible, incorporating minor variations in the training data that are likely to be noise rather than underlying patterns.
- An overfit model has small training errors, but may predict poorly. In essence, it has memorised the training set and cannot generalise well to the unseen data.
- Overfitting can be explained theoretically using bias-variance decomposition, which will be discussed later.
- Techniques dealing with overfitting will be discussed throughout the course.

Parametric vs nonparametric methods

- In **parametric** methods: we first assume a parametric form for the underlying model that generated the data, then estimate the parameters. The underlying model that generated the data is described by a functional form that depends on a vector of unknown parameters θ .
- E.g., simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

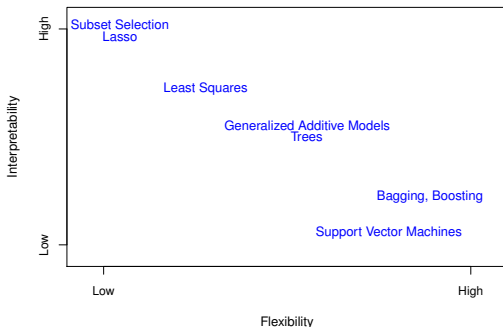
is a parametric model as we assume the model that generated data y_i , given x_i , is normal distribution with mean $\beta_0 + \beta_1 x_i$ and variance σ^2 . The set of unknown parameters is $\theta = (\beta_0, \beta_1, \sigma^2)$.

Parametric vs nonparametric methods

- In **nonparametric methods**, we don't assume a functional form for the underlying model that generated the data. An example is the K-nearest neighbours method, which we will study later.
- Parametric methods are faster to use, and more interpretable, but have the disadvantage of making stronger assumptions about the data.
- Nonparametric methods are more flexible, but have larger variance and can be computationally infeasible for large datasets.

Prediction accuracy vs interpretability

Particularly in business/economic applications, interpretability is an important consideration in addition to prediction accuracy. Highly flexible, nonparametric methods, tend to be less interpretable than simpler methods.



No free lunch theorem

All models are wrong, but some are useful. – George Box

- The field of machine learning proposes a large range of models and algorithms to solve supervised and unsupervised learning problems.
- However, there is no single model or approach that works optimally for all problems. This is sometimes called the **no free lunch theorem**.
- Therefore, applied statistical learning requires awareness of speed-accuracy-complexity trade-offs and data-driven consideration of different approaches for every problem.

Study guide

Study guide

- Recall three important concepts from these slides, and explain them in your own words.
- Use the review questions in the next slide to self-test on key concepts.
- Study the mathematical details in the module notes.
- Study (or revise) Chapters 1 and 2 of ISL. Reader Chapter 3 before the next lecture.

Review questions (1/2)

- What is predictive modelling?
- What is the difference between cross-sectional prediction and time-series forecasting?
- What is supervised learning?
- What is a loss function?
- What do we learn from statistical decision theory?

Review questions (2/2)

- What is overfitting?
- What is the difference between prediction and inference?
- What is the difference between parametric and nonparametric models? What are the advantages and disadvantages of each approach?